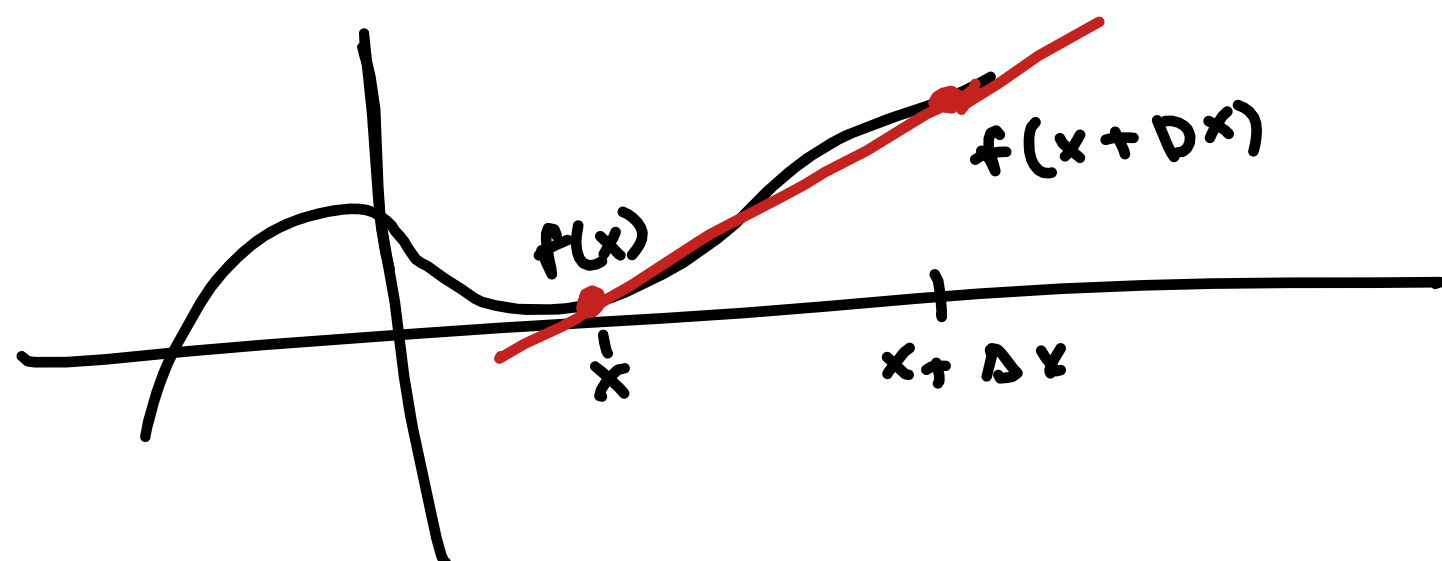


Definition of f'

Derivative

8/21/25

Derivative - Slope at an instantaneous point



Derivative: $\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$

alternate form

$$\lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c}$$

Notation

$f(x)$ - Equation of y ,

$f(a)$ - y @ a

derivatives $f'(x)$ - Equation of slopes

$f'(a)$ - Slope @ a

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$f'(c) = \lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c}$$

Example

A) If $f(x) = x^3$, $f'(x) = ?$

1) $\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ $f(x+h) = (x+h)^3$

2) $\lim_{h \rightarrow 0} \frac{(x+h)^3 - (x^3)}{h}$

3) $\lim_{h \rightarrow 0} \frac{x^3 + 3x^2h + 3xh^2 + h^3 - x^3}{h} = 3x^2$

4) $f'(x) = 3x^2$

B) equation of tangent line when $x=2$

1) $y - y_1 = m(x - x_1)$

$x=2$

$f(2) = 8$ [$x^3 \rightarrow 2^3$]

2) $y - 8 = 12(x - 2)$

$f'(2) = 12$ [$3x^2 \rightarrow 3(2)^2$]